ELISA Analysis & Decision-Support Workflow

Portfolio Case Study | SQL • Python • Power BI • Scientific Data Analysis

Purpose	**To demonstrate how multi-plate ELISA data, specimen and run metadata, predefined assay requirements and statistical evidence can be organized into a traceable workflow for scientifically defensible review and decision support.**

Synthetic portfolio dataset. QC thresholds are project assumptions used to demonstrate workflow logic; they are not presented as universal ELISA specifications.

1. What is ELISA?

ELISA (enzyme-linked immunosorbent assay) is a plate-based immunoassay that uses antibody-antigen recognition to detect or quantify a target. Depending on assay format, an antigen or capture antibody is immobilized on the plate, the target binds to its recognition partner, and an enzyme-linked detection step produces a measurable optical-density (OD) signal after substrate addition. In quantitative assays, the signal is interpreted against a calibration curve.

Capture antibody on plate	→	**Target antigen binds**	→	**Detection antibody binds target**	→	**Enzyme-linked detection**	→	**Substrate produces OD signal**

Simplified sandwich ELISA shown for illustration; the binding architecture differs across ELISA formats.

Common ELISA formats

Format	Principle	Typical use / implication
Direct	Immobilized antigen is detected by an enzyme-labelled primary antibody.	Simple detection with few reagents; limited signal amplification.
Indirect	Antigen is recognized by an unlabeled primary antibody and an enzyme-labelled secondary antibody.	Common for antibody detection/research; secondary reagent can amplify signal.
Sandwich	A capture antibody binds the analyte and a second detection antibody recognizes another epitope.	Widely used for specific quantification of proteins/biomarkers in complex matrices.

Competitive	Sample analyte competes with a labelled or plate-bound component; signal is typically inversely related to analyte amount.	Useful for small analytes or where a two-antibody sandwich is impractical.

2. Where is ELISA used, and who uses the results?

ELISA is widely used in pharmaceutical and biotechnology R&D, bioanalytical and contract research laboratories, clinical and diagnostic testing, academic research, and selected food, veterinary and public-health applications. The analyte and assay format vary by application, but reliable interpretation depends on acceptable assay performance and traceable data.

Stakeholders	What they need
Scientists and laboratory teams	Confidence that the assay performed correctly, results meet predefined requirements, and failed or unusual results can be investigated efficiently.
Quality teams and laboratory management	Consistent application of assay requirements, traceability of failures/retests, and visibility of recurring performance issues.
Study teams and data/digital systems teams	Reliable results for downstream scientific decisions and well-structured data that can move between analytical datasets, laboratory systems and reporting tools.

3. Core assay-validity concepts

Validity area	What it asks
Specificity / selectivity	Does the antibody-based assay measure the intended analyte with acceptable interference from matrix or related components?
Controls & background	Did the analytical run behave as expected, including blank, negative and positive-control response?
Precision	How reproducible are replicate measurements? CV% expresses variation relative to the mean.
Calibration / quantitative range	Does the standard response behave appropriately across concentration and support quantitative interpretation?
Accuracy / recovery	How close is the measured response or concentration to an expected or fortified value?
Sensitivity	Can sufficiently low analyte levels be reliably distinguished and quantified?

Robustness / spatial effects	Is assay performance sensitive to routine analytical conditions or plate position?

This portfolio project evaluates a decision-relevant subset of these concepts rather than claiming complete assay validation.

4. What problem is being addressed?

As ELISA studies expand across multiple plates, specimens, runs and study conditions, the challenge is not simply calculating OD or CV values. The analytical problem is to keep the data correctly connected and then determine what conclusions the evidence can legitimately support.

- **Data and traceability complexity** — well-level results must remain linked to specimens, plates, runs, reagent lots and execution metadata. Larger study cohorts, repeated sampling, repeat testing and limited specimen capacity per plate increase this complexity.
- **Scientific decision complexity** — a good calibration curve does not by itself establish plate validity, and an acceptable specimen CV cannot rescue a failed plate. Controls, calibrators and specimen precision must be interpreted in the correct sequence.
- **Investigation without over-interpretation** — differences by receipt delay, reagent lot, operator, instrument or plate position may identify useful investigation signals, but observational patterns do not automatically establish root cause.
- **Communication and follow-up** — reviewers need a concise way to see overall assay status, identify exceptions and trace affected specimen records without recalculating the analysis manually.

At a glance: what was addressed and how

1 TRACEABILITY	2 ASSAY TRUST	3 EVIDENCE	4 DECISION SUPPORT
Disconnected wells, specimens, plates, runs and metadata	A specimen result is not interpretable until the analytical plate is valid	Descriptive differences can be mistaken for causes	Reviewers need a concise view of exceptions and affected records
Relational data model + six SQL integrity checks	Controls + calibrators + 4PL + specimen replicate QC	Factor analysis + statistical testing + explicit limits on inference	Power BI summary + filters + specimen-level traceability

Problem → analytical response. The workflow first establishes trustworthy data and assay validity, then evaluates evidence and communicates decisions.

5. How are these problems addressed in this project?

The project uses a six-stage analysis and decision-support workflow. Project-specific implementation details are grouped here rather than mixed into the general ELISA background.

- **1. Data integrity & traceability** — connect wells → specimens → plates → runs → execution metadata. SQL checks well uniqueness, specimen/run referential integrity, expected replicate structure, plate-run consistency and completeness of critical run metadata. All six checks returned zero exceptions.
- **2. Assay run / plate validity** — apply predefined criteria to blanks, controls and calibrators before interpreting specimens. Calibration behaviour is assessed using 4PL modelling. Plates failing assay-level QC are excluded from downstream specimen-performance analysis.
- **3. Specimen-level QC** — calculate duplicate precision using CV% and classify results consistently as Pass, Review or Fail. Plate failure overrides an otherwise acceptable specimen replicate result.
- **4. Performance-factor investigation** — on QC-valid plates, examine matrix, receipt delay, reagent lot, collection site and relevant operational metadata. Exploratory plate-position analysis is retained only as a limited investigation because the assay layout was not designed as a formal edge-effect study.
- **5. Statistical evidence & interpretation** — use appropriate statistical tests to determine whether descriptive differences are supported by the available evidence. Separate observation, statistical association and causal interpretation.
- **6. Decision support & monitoring** — summarize assay outcomes and selected investigation factors in Power BI so reviewers can identify exceptions, examine subsets and trace individual specimen results.

Analytical data-integrity boundary

The SQL checks establish that the analytical dataset is structurally consistent for downstream analysis. They do not constitute formal computerized-system validation or a complete regulated audit-readiness assessment, which would additionally address areas such as user access, audit trails, change control, system testing, electronic-record governance, backup/recovery and approved procedures.

6. Analysis and key findings

Question	Evidence	Interpretation / decision
Data integrity & traceability	Six SQL integrity checks returned zero exceptions.	Analysis-ready relationships and critical metadata were structurally consistent for the intended portfolio analysis.
Assay / plate validity	PLT-26-046 and PLT-26-048 failed plate-control QC; calibrator QC passed across plates. 4PL fits were strong (R^2 0.9993–0.99997 with low RMSE).	Calibration quality alone does not establish plate validity. Specimens from failed plates inherit plate-failure status and are excluded from downstream factor analysis.
Specimen QC	Among 144 specimen test events on QC-valid plates: 126 Pass (87.5%), 7 Review (4.9%), 11 Fail (7.6%).	Most valid-plate specimen replicates met the predefined precision requirement; exceptions remain visible for review.
Matrix	Serum Pass 88.9%; plasma Pass 83.3%. Mann–Whitney p=0.236.	No statistical evidence of a meaningful CV difference between matrices in this dataset.
Receipt delay	Fail rate increased descriptively from 3.3% (0 d) to 6.7% (1 d) to 15.8% (2 d). Kruskal–Wallis p≈0.079; QC-outcome χ^2 p≈0.176.	A monitoring/investigation signal, but insufficient evidence to claim a receipt-delay effect.
Reagent lot	QC-concern rate 15.6% for RK26A17 vs 6.25% for RK26B04; Fisher exact p≈0.180.	Descriptive lot difference only; no root-cause claim, particularly given run-level confounding.
Operator	QC-concern rates 9.7%, 16.7% and 12.5%; χ^2 p≈0.53.	No evidence of operator-associated QC performance.
Plate position	Exploratory heat map showed isolated higher-CV positions but no consistent row, column or boundary pattern.	No edge-effect claim. A dedicated plate-layout study would be required for formal assessment.

7. Decision-support dashboard

A Power BI dashboard summarizes assay outcomes and provides interactive views of specimen QC, plate status and selected factors such as receipt delay, reagent lot, matrix, operator and instrument. Filters allow reviewers to examine affected subsets and trace individual specimen results. The dashboard is intentionally used as a communication and monitoring layer; the underlying SQL and Python analyses establish the QC logic and statistical evidence.

Dashboard view	What it supports

Overall specimen and plate status	Quick assessment of Pass / Review / Fail counts and failed plates.
Factor views and slicers	Exploration of selected specimen and process factors without treating descriptive differences as causal evidence.
Specimen-level table	Traceability from summary metrics back to individual specimen, plate and run records.

8. Conclusion and future directions

The project demonstrates how scientific assay knowledge can be translated into a structured data and analytical workflow: establish traceability first, determine assay validity before specimen interpretation, investigate potential performance signals only on valid analytical data, and use statistical evidence to control the strength of the conclusion. The result is a reproducible decision-support approach rather than a collection of disconnected analyses.

Future directions

- **Scale and automate the workflow** — integrate direct data ingestion from plate readers or laboratory systems and maintain traceable retest and QC-rule histories.
- **Extend assay-performance assessment** — use purpose-designed datasets to evaluate additional parameters such as recovery, sensitivity, robustness and spatial effects.
- **Develop advanced monitoring and investigation support** — use larger historical datasets for statistical process monitoring and, where appropriate, AI-assisted identification and investigation of recurring QC patterns.

Technical implementation and code

The report focuses on the scientific problem, analytical reasoning and decision implications. Reproducible implementation is maintained separately in the companion GitHub repository: SQL scripts for data-integrity, traceability and rule-based QC; a Python notebook for 4PL fitting, statistical testing and exploratory plate-position analysis; the Power BI file; synthetic data; and supporting documentation.

GitHub companion repository: [INSERT GITHUB LINK]